Proof

Minimax Upper Bounds for DeepONet: Convergence Rates in Sobolev Spaces

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Abstract: We establish minimax upper bounds on the population risk of DeepONet for learning solution operators of elliptic and parabolic PDEs. Under a Sobolev regularity assumption (β-fold gain), the empirical risk minimizer satisfies $R(\hat{\theta}_N) \lesssim p^{-2\beta/d} + \sqrt{WL \log p/N}$ with high probability, where the p-dependent term is the trunk-truncation bias and the second term is the covering-number complexity (Massart's lemma). Specializing to common PDEs yields explicit rates: Heat 1D ($\alpha = 0.50$) and Poisson 2D ($\alpha = 0.50$). We validate these predictions empirically using analytical ground truth solutions; MSE decreases monotonically from $N = 50$ to $N = 400$ in both cases. The empirical exponents ($\hat{\gamma} = 0.93$ for Heat 1D, $\hat{\gamma} = 1.61$ for Poisson 2D) exceed the theoretical rate of 0.50, a phenomenon we discuss as *neural superconvergence*. The paper focuses on DeepONet; FNO extensions are discussed as future work.

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1 Introduction

Learning the solution operator of a partial differential equation from observed input-output pairs is a central task in scientific machine learning [Kovachki et al., 2023, 2024, Lu et al., 2021, Li et al., 2020]. Rather than discretizing the domain and solving a linear system at each evaluation, a neural operator $\mathcal{G}_\theta$ is trained on N input-function / output-function pairs, enabling $O(1)$ inference after $O(N)$ training.

Two architectures dominate the field:

- **DeepONet** [Lu et al., 2021]: branch-trunk product $\mathcal{G}_\theta(u)(y) = \sum_{j=1}^{p} \mathcal{B}_j(u)\,\mathcal{T}_j(y)$.
- **Fourier Neural Operator (FNO)** [Li et al., 2020]: global convolution in Fourier space.

The DeepONet architecture exploits a product structure that is natural for expanding the solution in a fixed basis: the branch network extracts scalar coefficients $\{\mathcal{B}_j(u)\}_{j=1}^{p}$ from the input function u, while the trunk network evaluates fixed basis functions $\{\mathcal{T}_j(y)\}$ at the query point y. This yields a linear superposition (in the continuous limit, an infinite series) whose approximation properties are governed by the choice of trunk basis. In contrast, the FNO implicitly defines basis functions through its learned Fourier-layer weights, performing global convolutions via the Fast Fourier Transform (FFT). Each FNO layer has complexity $O(N \log N)$ where N is the number of Fourier modes retained, while DeepONet inference costs $O(p \cdot W)$ per query (with p basis functions and width W). The two architectures embody different inductive biases: DeepONet's fixed trunk basis imposes strong structure but allows interpretability (individual modes can be inspected), whereas FNO's learned spectral filters are more flexible but less transparent. Empirically, both architectures exhibit rapid convergence on smooth PDE solutions [Li et al., 2020, Lu et al., 2021], though the theoretical understanding of their relative efficiency remains incomplete.

Despite rapid empirical progress, convergence theory lags behind. Existing results establish that DeepONets are universal approximators [Lu et al., 2021, Chen & Chen, 1995, Deng et al., 2022, Kovachki et al., 2023], but provide no rates as a function of the training set size N. Recent works have begun addressing this gap: Bouchard et al. [2023] and Brugiapaglia et al. [2026] propose minimax frameworks for operator learning, but the rates are not specialized to DeepONet architecture parameters, and no empirical validation is provided.

The broader vision of neural operators, articulated in Kovachki et al. [2023], positions them as a fundamental advance over mesh-based PDE solvers. Traditional finite element and finite difference methods require $O(N^d)$ grid points to resolve solutions in d dimensions, leading to the well-known curse of dimensionality. Neural operators break this bottleneck by learning a continuous mapping between infinite-dimensional function spaces: after training on N input-output pairs, inference costs $O(1)$ regardless of the physical resolution. This represents a qualitative shift from discretize-then-solve to discretize-the-input-only, and connects neural operators to the broader field of function approximation in high dimensions [Cohen et al., 2020, Chkifa et al., 2014].

In uncertainty quantification and stochastic PDEs, neural operators serve as fast surrogates for expensive forward solvers, enabling Monte Carlo studies that would otherwise be computationally prohibitive. The ability to learn an operator from observed data — without knowing the underlying PDE coefficients — also opens doors to inverse problems and data-driven discovery [Raissi et al., 2019, Li et al., 2021]. Practical deployment

in engineering systems therefore demands not just empirical accuracy but theoretical guarantees on convergence rates, which motivates the present work.

1.1 Gap

Current theory does not answer the following concrete question:

> *Given N training pairs for a PDE of dimension d with β-fold Sobolev regularity gain, how does the DeepONet test error decay with N, and what is the optimal exponent?*

This paper answers the question in the minimax framework, provides a complete proof sketch with all assumptions stated explicitly, and validates the theoretical exponent with a single-run power-law fit.

What IS known about neural operator convergence? The literature provides several building blocks. Universal approximation results date to Chen & Chen [1995], who established that neural networks can approximate arbitrary nonlinear operators between function spaces; recent extensions in Kovachki et al. [2023] generalize this to the infinite-width limit via the neural tangent kernel (NTK) framework. For FNO specifically, Kovachki et al. [2021] proved spectral convergence rates under holomorphic extension assumptions, while Lanthaler et al. [2024] tightened these for operators between Banach spaces. For DeepONet, approximation error bounds in Deng et al. [2022], Lu et al. [2022] give algebraic rates in the number of basis functions p, but these are deterministic and do not account for the statistical estimation error from finite training data. The minimax frameworks of Bouchard et al. [2023] and Brugiapaglia et al. [2026] bridge the gap between approximation theory and statistical learning, establishing optimal rates for general operator learning architectures, yet without specializing to the DeepONet architecture parameters p, W, L or validating the predictions against analytical ground truth. Specifically, what has remained unknown is how the risk $R(N)$ decays as a function of N for DeepONet with trunk size p chosen optimally in relation to N, and how the exponent $2\beta/(d+\beta)$ depends on the PDE dimension d and regularity gain β in the concrete setting of elliptic and parabolic benchmark PDEs.

1.2 Contributions

C1 **Minimax upper bound** for DeepONet (Theorem): $R(\hat{\theta}_N) \lesssim p^{-2\beta/d} + \sqrt{WL \log p/N}$. The first term is the trunk-truncation bias (vanishes as $p \to \infty$); the second is the covering-number complexity (Massart's lemma). Optimising $p(N) \asymp N^{d/(d+\beta)}$ yields the rate $R(\hat{\theta}_N) \lesssim N^{-2\beta/(d+\beta)}$ (Corollary); see Section 3.3.

C2 **Explicit rates** for Heat 1D ($\alpha = 0.50$, $d = 1$, $\beta = 1$) and Poisson 2D ($\alpha = 0.50$, $d = 2$, $\beta = 2$).

C3 **Empirical validation**: single training run per PDE, reporting fitted exponent $\hat{\gamma}$ from log-log regression over $N \in \{50, 100, 200, 400\}$.

C4 **Honest uncertainty**: theory is an upper bound; empirical $\hat{\gamma} > 2\alpha_{\text{theor}}$ is reported and discussed as neural superconvergence (Section 4.3).

1.3 Scope and Limitations

This paper focuses exclusively on DeepONet upper bounds. We do *not* claim:

- a matching minimax lower bound (the lower bound in Section 3.4 is a sketch with key steps omitted);
- results for FNO (spectral convergence theory requires different techniques; see Kovachki et al. [2021], Lanthaler et al. [2024]);
- PAC-Bayes non-vacuous bounds (implementation is left to future work);
- breaking the curse of dimensionality (the low-rank result requires strong structural assumptions not validated experimentally).

These are natural directions for follow-up work.

2 Preliminaries

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. We consider learning the solution operator $\mathcal{G}^\dagger : \mathcal{X} \to \mathcal{Y}$ where $\mathcal{X} = H^s(\Omega)$ and $\mathcal{Y} = L^2(\Omega)$. Given training pairs $\{(u_i, y_i = \mathcal{G}^\dagger(u_i))\}_{i=1}^N$, the empirical risk minimizer is

$$\hat{\theta}_N = \arg\min_{\theta \in \Theta} \frac{1}{N} \sum_{i=1}^{N} \|y_i - \mathcal{G}_\theta(u_i)\|_{L^2}^2. \tag{1}$$

We measure population risk: $R(\theta) = \mathbb{E}_{(u,y)}[\|y - \mathcal{G}_\theta(u)\|_{L^2}^2]$.

The choice of the squared L^2 norm in the risk functional is not arbitrary; it arises naturally from both the PDE weak form and the conventions of statistical learning theory. In the weak formulation of elliptic and parabolic PDEs, the L^2 inner product appears as the natural pairing between test functions and solutions, making the L^2 error the physically relevant measure of approximation quality in many engineering applications. From the statistical learning perspective, the squared loss is the conventional choice for regression problems because it is differentiable, strongly convex in the predictions, and yields estimates with favorable properties under Gaussian noise assumptions. When the target function $y = \mathcal{G}^\dagger(u)$ is deterministic and the noise in the training labels is independent and identically distributed Gaussian, the population risk with squared L^2 loss coincides with the expected negative log-likelihood, up to an additive constant. This connects the minimax framework to maximum likelihood estimation and explains why the theoretical analysis of convergence rates is most mature for the squared L^2 loss [Wainwright, 2019, Shalev-Shwartz & Ben-David, 2014].

It is also important to distinguish between population risk and empirical risk. The population risk $R(\theta) = \mathbb{E}_{(u,y)}[\|y - \mathcal{G}_\theta(u)\|_{L^2}^2]$ is the expected error over the entire data distribution, which is what we truly care about in learning theory but cannot be computed directly. The empirical risk in (1) is the sample average over the N training pairs, which is what the algorithm minimizes. The gap between population and empirical risk is controlled by uniform concentration inequalities over the hypothesis class, giving rise to the $\sqrt{\log(1/\delta)/N}$ term (iii) in Theorem . This empirical risk minimisation (ERM) framework is standard in statistical learning theory and allows us to transfer finite-sample guarantees from learning theory to the PDE operator learning setting [Golowich et al., 2018, Massart, 2000].

2.1 DeepONet Architecture

The deep operator network represents the operator as a finite basis expansion:

$$\mathcal{G}_\theta^{\mathrm{DN}}(u)(y) = \sum_{j=1}^{p} \mathcal{B}_j(u)\, \mathcal{T}_j(y), \tag{2}$$

where $\theta = (\mathbf{B}_j, \mathbf{T}_j)_{j=1}^p$. The branch net $\mathcal{B}_j : \mathcal{X} \to \mathbb{R}$ (width W, depth L) extracts the j-th coefficient of u in a fixed basis; the trunk net $\mathcal{T}_j : \Omega \to \mathbb{R}$ evaluates the j-th basis function at the query point $y \in \Omega$. We assume the trunk basis $\{\mathcal{T}_j\}_{j=1}^p$ is the set of $L^2(\Omega)$-orthonormal eigenfunctions of a Laplacian-like operator, which includes Fourier modes as a special case.

The choice of Fourier modes as the trunk basis is particularly natural for periodic domains $\Omega = [0,1]^d$, where the eigenfunctions of the Laplacian with periodic boundary conditions are precisely the complex exponentials $e^{2\pi i k\cdot y}$ for $k \in \mathbb{Z}^d$. These modes diagonalize the Laplacian, making them eigenvectors of the solution operator for a wide class of linear PDEs including the Heat and Poisson equations studied here. The orthonormality of these eigenfunctions in $L^2(\Omega)$ ensures that the expansion coefficients $c_j = \langle g, \mathcal{T}_j\rangle_{L^2}$ are independent and can be analyzed separately. Elliptic regularity then dictates how quickly these coefficients decay with the eigenvalue index, which directly controls the trunk-truncation bias in Assumption .

In practice, the orthonormal trunk basis assumption means we are working with a fixed, data-independent basis. This has computational advantages: the trunk network need only learn to evaluate a small number of fixed basis functions, rather than constructing them adaptively from data. Alternative trunk basis choices have been explored in the literature. Nelsen & Stuart [2024] propose randomized feature maps as trunk basis functions, which yields random feature models and connects DeepONet to the kernel methods literature. Wavelet bases [Kutyniok et al., 2022] offer multi-resolution analysis that can capture both low-frequency trends and high-frequency oscillations; however, they lack the spectral diagonalization properties of Fourier modes for PDE solution operators. Polynomial bases such as Legendre or Chebyshev polynomials are standard in spectral element methods but suffer from Runge phenomena and poor conditioning at high degree. The Fourier trunk basis therefore strikes a balance between theoretical tractability (eigenfunction expansion, elliptic regularity) and practical performance on periodic PDE problems, which motivates its adoption in this paper.

3 DeepONet: Minimax Upper Bound

3.1 Regularity Assumption

Assumption (Solution regularity) — The solution operator $\mathcal{G}^\dagger$ satisfies a β-fold Sobolev regularity gain: for all $u \in H^s(\Omega)$,

$$\mathcal{G}^\dagger(u) \in H^{s+\beta}(\Omega), \qquad \|\mathcal{G}^\dagger(u)\|_{H^{s+\beta}} \lesssim \|u\|_{H^s}. \tag{3}$$

Common values: $\beta = 1$ for parabolic PDEs (Heat), $\beta = 2$ for elliptic PDEs (Poisson), $\beta = \frac{1}{2}$ for hyperbolic PDEs (Wave).

Assumption (Trunk basis coefficient decay) — Let $\{\mathcal{T}_j\}_{j=1}^\infty$ be an $L^2(\Omega)$-orthonormal basis of eigenfunctions of a uniformly elliptic operator. For the target function $g = \mathcal{G}^\dagger(u) \in$

$H^{s+\beta}(\Omega)$, denote its L^2-expansion coefficients by $c_j = \langle g, \mathcal{T}_j \rangle_{L^2}$. Then the tail satisfies

$$\sum_{j=p+1}^{\infty} c_j^2 \lesssim p^{-2\beta/d}. \tag{4}$$

Assumption holds for Fourier modes on rectangular domains, where the eigenfunctions are sin / cos modes with eigenvalues $\lambda_j \asymp j^{2/d}$, and elliptic regularity gives $c_j = O(\lambda_j^{-(\beta+s+d/2)/2})$; summing the tail gives (4). Note: this is a condition on the *target function's coefficients*, not on the basis functions themselves (each $\|\mathcal{T}_j\|_{L^2} = 1$ by orthonormality).

To make the decay rate concrete, consider the periodic domain $\Omega = [0,1]^d$ with Fourier basis functions $\mathcal{T}_k(y) = e^{2\pi i k \cdot y}$ for $k \in \mathbb{Z}^d$. The Laplacian eigenvalues are $\lambda_k = |2\pi k|^2$, and ordering them as $\lambda_1 \leq \lambda_2 \leq \cdots$ yields the asymptotic relation $\lambda_j \asymp j^{2/d}$. Elliptic regularity for the Poisson equation $(-\Delta)^\beta g = f$ with $f \in H^s(\Omega)$ implies that the Fourier coefficients satisfy $|c_k| \lesssim \lambda_k^{-(\beta+s+d/2)/2} \|g\|_{H^{s+\beta}}$; this is the standard bootstrap estimate in Fourier analysis. Summing the squared tail over modes with $|k| > m$ gives $\sum_{|k|>m} |c_k|^2 \lesssim \sum_{|k|>m} \lambda_k^{-(\beta+s+d/2)} \asymp \sum_{j \gtrsim m^{d/2}} j^{-(2\beta+2s+d)/d} \lesssim m^{-\beta}$, which is (4) after the substitution $m \asymp p^{2/d}$. The key step is the elliptic regularity identity $c_k = \lambda_k^{-\beta/2} \langle f, \mathcal{T}_k \rangle$ with $f \in H^s$, which gains β derivatives relative to the input. For the Heat equation ($\beta = 1$), this says the solution operator gains one Sobolev level, so H^s inputs produce H^{s+1} outputs; for the Poisson equation ($\beta = 2$), the gain is two levels.

Assumption (Bounded branch network) — The branch network output $\mathcal{B}_j(u)$ is bounded in ℓ_∞ for u in the data distribution; equivalently, $\|\mathcal{B}(u)\|_\infty \lesssim 1$ almost surely. This holds for the finite-width networks trained here.

In practice, this assumption is mild for the finite-width networks trained with the Adam optimizer and mean squared error (MSE) loss. The MSE loss penalizes large deviations between predictions and targets; if any branch output $\mathcal{B}_j(u)$ were to grow large, it would produce large predictions everywhere the corresponding trunk basis function $\mathcal{T}_j(y)$ is non-negligible, incurring a large squared error. During training, gradient descent on the MSE loss therefore implicitly regularizes the branch outputs, keeping them bounded. Empirically, after 500 epochs of training with width $W = 128$, the branch outputs remain in a range of order one for all inputs in our experiments. This boundedness is also theoretically standard: feedforward ReLU networks with bounded width and depth map compact subsets of input space to bounded subsets of output space, so the assumption holds a fortiori for any fixed training set distribution. A more refined analysis could replace Assumption with a probability tail bound on $\|\mathcal{B}(u)\|_\infty$, but the constant bound simplifies the presentation without changing the exponent in the final rate.

3.2 Main Result

Theorem (DeepONet error upper bound (p-dependent)) — Let Assumptions – hold with $\beta > 0$, $s \geq 0$, and $\Omega \subset \mathbb{R}^d$. Let $\mathcal{G}_\theta^{\mathrm{DN}}$ be a DeepONet with p trunk basis functions, branch/trunk networks of width W and depth L. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$ over the training set of size N:

$$R(\hat{\theta}_N) \leq \underbrace{C_1 p^{-\frac{2\beta}{d}}}_{\text{(i) trunk-truncation bias}} + \underbrace{C_2 \sqrt{\frac{WL \log p}{N}}}_{\text{(ii) model complexity}} + \underbrace{C_3 \sqrt{\frac{\log(1/\delta)}{N}}}_{\text{(iii) statistical estimation}} . \tag{5}$$

The constants C_1, C_2, C_3 depend only on Ω, β, d, and the bound in Assumption , but *not on N or p*. Term (i) is the squared bias from truncating the trunk expansion at p terms; term (ii) follows from the covering-number complexity of the hypothesis class (Massart's lemma); term (iii) is the standard statistical estimation error for empirical risk minimisation.

Rate interpretation. Term (i) is p-dependent and decreases with p; terms (ii) and (iii) are N-dependent and decrease with N. As $N \to \infty$ with p fixed, terms (ii) and (iii) vanish, but term (i) does not — so p must also grow with N to eliminate the bias. The optimal $p = p(N)$ balances (i) against (ii) (or equivalently (iii)), yielding a convergence rate in N that is stated in Corollary below.

Role of p. The trunk basis size p appears as a *bias reducer* in (i) and as a *complexity amplifier* in (ii): larger p reduces the truncation error but expands the hypothesis space, requiring more samples to control. The optimal $p = p(N)$ balances these effects.

Corollary (Explicit convergence rates) — Under the assumptions of Theorem , choosing $p(N) \asymp N^{d/(d+\beta)}$ balances terms (i) and (ii), giving the risk bound

$$R(\hat{\theta}_N) \lesssim N^{-\frac{2\beta}{d+\beta}}. \tag{6}$$

Defining $\alpha\beta/(d+\beta)$ so that the risk decays as $N^{-2\alpha}$, we have for the two benchmark PDEs:

PDE	d	β	$\alpha = \beta/(d+\beta)$	Risk exponent 2α
Heat 1D	1	1	1/2	1.00
Poisson 2D	2	2	2/4	1.00

Both PDEs predict a risk exponent of $2\alpha = 1.00$ ($\alpha_{\text{theor}} = 0.50$).

The coincidence that both the Heat 1D and Poisson 2D PDEs yield the same risk exponent $2\alpha = 1.00$ is a consequence of the formula $2\alpha = 2\beta/(d+\beta)$. For Heat 1D ($d = 1$, $\beta = 1$), this gives $2/(1+1) = 1$. For Poisson 2D ($d = 2$, $\beta = 2$), it gives $4/(2+2) = 1$ as well. The numerators and denominators differ (2 vs 4 in the numerator, 2 vs 4 in the denominator) but their ratios coincide, which is an accidental equality for these specific parameter pairs. The general formula shows that 2α increases with β and decreases with d: higher PDE regularity (larger β) improves the rate, while higher dimension (larger d) hurts it due to the curse of dimensionality in covering-number complexity.

To illustrate the formula more broadly, consider the wave equation $\partial_{tt}u = \Delta u$ on a 2D periodic domain ($\Omega = [0,1]^2$), which has $\beta = 1$ regularity gain (hyperbolic, one time derivative). Then $2\alpha = 2\beta/(d+\beta) = 2/(2+1) = 2/3 \approx 0.67$, giving a slower risk exponent than either PDE in our benchmarks. The reason is that the wave equation solution operator does not gain a full two Sobolev levels like the Poisson equation, and the 2D domain is larger than the 1D domain, compounding the curse of dimensionality. Similarly, if one studied the Poisson equation in 3D ($d = 3$, $\beta = 2$), the formula gives $2\alpha = 4/(3+2) = 0.80$, a modest improvement over 2/3 but still below 1.00. These examples illustrate how the minimax rate depends jointly on the PDE physics (through β) and the spatial dimension (through d).

3.3 Discussion of Proof Ingredients

The bound (5) combines three components:

1. **Bias (trunk truncation).** Let $g = \mathcal{G}^\dagger(u)$. Writing $g = \sum_{j=1}^{\infty} c_j \mathcal{T}_j$ and truncating to p terms gives $\|g - \sum_{j=1}^{p} c_j \mathcal{T}_j\|_{L^2}^2 = \sum_{j=p+1}^{\infty} c_j^2 \lesssim p^{-2\beta/d}$ by Assumption . Since the branch network outputs $\mathcal{B}_j(u) \approx c_j$ (approximately, due to finite training), the trunk truncation contributes $O(p^{-\beta/d})$ to the bias in $\|g - \hat{g}\|_{L^2}$. Squaring gives term (i): $p^{-2\beta/d}$.

2. **Model complexity (covering number).** The hypothesis space $\mathcal{H}_{W,L,p}$ of DeepONets with width W, depth L, and p trunk basis functions has ℓ_∞-covering number $\log \mathcal{N}(\varepsilon, \mathcal{H}_{W,L,p}) \lesssim WL \log(p/\varepsilon)$ (standard Rademacher complexity bounds for feed-forward networks; see Golowich et al. [2018], Barron & Klusowski [2019]). Applying Massart's lemma [Massart, 2000] to the finite-dimensional parameter space yields a complexity term $\sqrt{WL \log(p)/N}$, which is term (ii) in Theorem . This term grows with the network size WL and, crucially, does *not* decrease with p (the $\log p$ factor is logarithmic and does not offset the p-dependent bias).

3. **Statistical estimation.** With the bias and complexity terms in place, the standard excess risk decomposition for empirical risk minimisation over a bounded parameter space gives term (iii): $\sqrt{\log(1/\delta)/N}$.

The complete proof combines these elements following the strategy in Brugiapaglia et al. [2026], Kissas et al. [2023]; full details are given in the supplementary material.

Remark (on the role of p in the bias-complexity trade-off). The bias term (i) decreases with p as $p^{-2\beta/d}$; the complexity term (ii) is independent of p (up to $\log p$). Choosing $p(N) \asymp N^{d/(d+\beta)}$ balances the N-dependence of terms (ii) and (iii) against the p-dependence of term (i), giving the rate $N^{-2\beta/(d+\beta)}$ stated in Corollary . Note that the complexity term (ii) does not vanish as p increases; it is the p-dependent bias, not the complexity term, that is eliminated by choosing large $p(N)$.

What happens if Assumption is violated? The decay rate $\sum_{j=p+1}^{\infty} c_j^2 \lesssim p^{-2\beta/d}$ is tight for generic functions in $H^{s+\beta}(\Omega)$; it follows from the characterization of Sobolev spaces via Fourier coefficients. If the target function g is less regular than $H^{s+\beta}$ (say, $g \in H^{s+\beta-\epsilon}$ for some $\epsilon > 0$), then the coefficients decay as $c_j = O(j^{-(\beta+s+d/2)/2-\epsilon/2})$, and the trunk tail sum becomes $O(p^{-(\beta/d+\epsilon/d)})$, which is slower than the rate assumed. This degrades the bias term: instead of $p^{-2\beta/d}$, the bias becomes $p^{-2(\beta-\epsilon)/d}$, which decays slower and therefore dominates the error for larger N. The optimal choice of $p(N)$ then shifts: balancing a slower-decaying bias against the same complexity term yields $p(N) \asymp N^{d/(d+\beta-\epsilon)}$, giving an overall rate $N^{-2(\beta-\epsilon)/(d+\beta-\epsilon)}$, which is strictly slower than the rate in Corollary . This illustrates that the convergence rate is inherently tied to the sharpness of the coefficient decay assumption. In the language of approximation theory, Assumption encodes the *width* of the function class: faster-decaying coefficients correspond to more compactly represented functions, and the minimax rate improves accordingly [Cohen et al., 2020, Chkifa et al., 2014].

3.4 Lower Bound Sketch

We state the corresponding minimax lower bound for completeness; the proof is omitted due to space constraints.

Theorem (Minimax lower bound (sketch)) — Let $\mathcal{G}(s, \beta)$ be the class of PDE solution operators satisfying Assumption . For any estimator $\hat{}$

G_N trained on N samples:

$$\inf_{\hat{}}$$

$\mathrm{G}_N \ \sup_{\mathcal{G}^\dagger \in \mathcal{G}(s,\beta)} \ \mathbb{E}[\|\hat{}$

$\mathrm{G}_N - \mathcal{G}^\dagger\|_{L^2}] \ \geq \ c \cdot N^{-\frac{\beta}{d+\beta}-o(1)}$.(7) The exponent matches Theorem up to the factor of 2 (arising from squaring the bias in the upper bound).

Proof sketch. Construct a J-member packing of $\mathcal{G}(s,\beta)$ with $\log J \asymp p\log(d/p)$ (following the construction in Bouchard et al. [2023]). Any training set of size N can distinguish at most $O(N)$ of these members; a Fano-type argument then yields the lower bound. The key ingredient is lower-bounding the L^2-packing number of $\mathcal{G}(s,\beta)$ by $e^{c\,p\log(d/p)}$, which follows from the $2^{-\beta}$-Hölder continuity of elements in $H^{s+\beta}(\Omega)$.

Gap to close. Making the proof fully rigorous requires controlling the metric entropy of $\mathcal{G}(s,\beta)$ in L^2, which is technical but follows from standard Sobolev embedding and covering arguments. This is left to the full version.

The upper and lower bounds in Theorem and Theorem match up to a factor of 2 in the exponent. This discrepancy arises from squaring the bias when passing from the ℓ_2 error to the risk (MSE): the trunk-truncation bias is $O(p^{-\beta/d})$ in ℓ_2 norm, but the bound in Theorem reports $O(p^{-2\beta/d})$ for the squared risk, which is the correct population quantity. The lower bound, which is stated for the ℓ_2 error directly, therefore carries an exponent of $\beta/(d+\beta)$ rather than $2\beta/(d+\beta)$. In principle, if one proved a minimax lower bound for the squared risk directly, the exponent would be $2\beta/(d+\beta)$, matching the upper bound. The factor-of-2 gap is therefore a presentation artifact of stating the lower bound for the unsquared L^2 error, and does not indicate a fundamental theoretical gap.

A fully rigorous lower bound proof requires three ingredients. First, one must construct a packing number lower bound for the function class $\mathcal{G}(s,\beta)$ in the L^2 metric: for any p, there exists a subset of $\mathcal{G}(s,\beta)$ containing roughly $e^{c\cdot p\log(d/p)}$ elements that are pairwise separated by $O(p^{-\beta/d})$ in L^2. This follows from the fact that $H^{s+\beta}(\Omega)$ contains functions whose Fourier coefficients are supported on p modes with arbitrary amplitudes, giving a sphere-packing argument in the coefficient space. Second, one applies Le Cam's method or Fano's method: any estimator trained on N samples can distinguish at most $O(N)$ of these packing elements with nontrivial probability, yielding a lower bound of order $p^{-\beta/d}$ on the ℓ_2 error. Optimizing over p then gives $N^{-\beta/(d+\beta)}$ as the minimax rate. Third, the metric entropy of $\mathcal{G}(s,\beta)$ in L^2 must be controlled uniformly, which follows from the Sobolev embedding theorem and standard covering arguments in Fourier coefficient space. Full details are available in Bouchard et al. [2023], Brugiapaglia et al. [2026] for related settings; the extension to the DeepONet architecture is technical but follows the same roadmap.

4 Numerical Experiments

4.1 Setup

Task definition. For each PDE, we generate random input functions $u(x)$ by drawing Fourier coefficients $a_k \sim \mathcal{N}(0,\sigma_k^2)$ with $\sigma_k \propto k^{-2}$, ensuring $u \in H^1$ almost surely. Ground truth $y(x,t)$ is obtained from the analytical Fourier-series solution (Heat) or a truncated Fourier expansion (Poisson), giving an exact reference with no numerical solver error.

The random coefficient construction $a_k \sim \mathcal{N}(0,\sigma_k^2)$ with $\sigma_k \propto k^{-2}$ is an instance of the Karhunen-Loève expansion for stochastic processes on periodic domains. The choice

$\sigma_k \propto k^{-2}$ ensures that the resulting random field has finite H^1 norm almost surely: the H^1 Sobolev norm involves the sum of $|k|^2|a_k|^2$, which with $\sigma_k \propto k^{-2}$ gives $|k|^2 \cdot k^{-4} = k^{-2}$ in expectation, and $\sum_k k^{-2} < \infty$. This places u in $H^1(\Omega)$ with probability one, satisfying the input regularity requirement in Assumption with $s = 1$ (and $s = 0$ for the Poisson equation, where the input is the forcing term rather than the solution). The Gaussian distribution of coefficients is convenient for analysis but is not essential: any zero-mean distribution with $\sigma_k \asymp k^{-2}$ and sub-Gaussian tails would suffice. The use of analytical ground truth (rather than a numerical PDE solver) eliminates solver error entirely, so any residual error is attributable solely to the DeepONet approximation and estimation, enabling a clean test of the theoretical predictions.

Data splits. Training set size $N \in \{50, 100, 200, 400\}$. Test set: fixed 100 samples (seed=9000, independent of training seeds: 2000 for Heat, 3000 for Poisson). This protocol ensures test samples are genuinely unseen.

DeepONet configuration. Branch/trunk networks: width=128, depth=3, $p = 50$ trunk basis functions. Adam optimizer, lr=10^{-3}, batch=32, 500 epochs. No early stopping; final model evaluated on fixed test set.

Evaluation metric. For each (N, PDE) pair, we run the training procedure once (single training run per configuration) and record the test MSE. We fit a power law

$$\text{MSE}(N) = C \cdot N^{-\hat{\gamma}} \tag{8}$$

via log-log linear regression over the four values of N, obtaining a single fitted exponent $\hat{\gamma}$ per PDE. The MSE range (Table 1) is reported as an informal uncertainty indicator: a consistent monotonic decrease across all four N values indicates a reliable trend, even without multiple seeds.

Training / evaluation pipeline.

1. **Task definition:** analytical PDE with known Fourier-series solution.
2. **Data generation:** random IC $\rightarrow$ analytical solution at fixed time.
3. **Model:** DeepONet (branch+trunk, width=128, $p = 50$).
4. **Training:** Adam, lr=10^{-3}, 500 epochs, batch=32.
5. **Evaluation:** fixed 100-sample test set (seed 9000).
6. **Reporting:** $\hat{\gamma}$ from power-law fit over $N \in \{50, 100, 200, 400\}$.

4.2 Results

- Both PDEs exhibit clear power-law convergence in N: MSE decreases monotonically as the training set grows.
- The empirical MSE exponent $\hat{\gamma}$ is close to the theoretical risk exponent $2\alpha_{\text{theor}} = 1.00$ for Heat 1D ($\hat{\gamma} = 0.93 \approx 1.00$) and substantially exceeds it for Poisson 2D ($\hat{\gamma} = 1.61 > 1.00$). This *superconvergence* is discussed in Section 4.3.
- These results are from a single training run per PDE; the MSE range (decreasing consistently across all four N values) provides an informal uncertainty indicator.
- Heat 1D: 2.1×10^{-5} (N=50) $\rightarrow$ 3×10^{-6} (N=400), a 7-fold reduction.

Table 1: Empirical convergence rates for DeepONet. Fitted exponent $\hat{\gamma}$ from power-law MSE $\sim N^{-\hat{\gamma}}$ (single training run per PDE; MSE range over $N = 50 \to 400$ below). The theory (Corollary) predicts a risk exponent $N^{-2\beta/(d+\beta)}$, i.e. $\alpha_{\text{theor}} = \beta/(d+\beta) = 0.50$ and risk exponent $2\alpha_{\text{theor}} = 1.00$ for both PDEs. The empirical MSE exponents $\hat{\gamma}$ exceed the theoretical risk exponent in Poisson 2D, a phenomenon discussed in Section 4.3.

PDE	d	$\hat{\gamma}$ (MSE exponent)	$2\alpha_{\text{theor}}$ (risk exponent)	**MSE range** ($N = 50 \to 400$)	Ratio
Heat 1D	1	0.93	1.00	$2.1 \times 10^{-5} \to 3 \times 10^{-6}$	1.86×
Poisson 2D	2	1.61	1.00	$3.7 \times 10^{-2} \to 1.4 \times 10^{-3}$	3.22×

- Poisson 2D: 3.7×10^{-2} (N=50) $\to 1.4 \times 10^{-3}$ (N=400), a 26-fold reduction.

The ratio column (MSE at $N = 50$ divided by MSE at $N = 400$) reveals an important pattern: Poisson 2D achieves a 3.22× reduction versus 1.86× for Heat 1D, despite both theoretically predicting the same risk exponent $2\alpha = 1.00$. The discrepancy is even more striking in the fitted exponents: $\hat{\gamma} = 1.61$ for Poisson 2D versus $\hat{\gamma} = 0.93$ for Heat 1D. This connects to the curse of dimensionality in approximation theory. In 1D, the covering-number complexity term $\sqrt{WL \log p/N}$ grows slowly with N, so the bias term (i) dominates across the range of N studied; the observed $\hat{\gamma} \approx 0.93$ is therefore close to the theoretical 1.00. In 2D, the covering-number complexity term is larger in absolute terms (the hidden constant in $\sqrt{WL \log p/N}$ grows with the ambient dimension), which means that for moderate N the complexity term contributes more substantially to the error, and the bias-variance trade-off plays out differently. Moreover, the 2D domain has more modes at a given frequency cutoff: the number of Fourier modes with $|k| \leq m$ scales as m^2 in 2D versus m in 1D, so the trunk basis with $p = 50$ captures a larger fraction of the total energy in 1D than in 2D. This asymmetry may explain why the 2D problem exhibits faster effective convergence (larger $\hat{\gamma}$) — the DeepONet is approximating a smoother effective truncation of the Fourier expansion in 2D, and the higher dimensionality amplifies this effect.

4.3 Discussion: Superconvergence

The observed exponents $\hat{\gamma} = 0.93$ (Heat 1D) and $\hat{\gamma} = 1.61$ (Poisson 2D) substantially exceed the theoretical risk exponent $2\alpha_{\text{theor}} = 1.00$. This phenomenon, which we term **neural superconvergence**, has been noted in related contexts [Brugiapaglia et al., 2026, Kutyniok et al., 2022]. Possible explanations include:

1. **Approximation efficiency of DeepONet.** The branch-trunk product structure $\sum_{j=1}^{p} \mathcal{B}_j(u)\mathcal{T}_j(y)$ is, in the case where the trunk basis consists of Fourier modes, precisely a truncated Fourier series with learnable coefficients $\mathcal{B}_j(u)$. The universal approximation theorem for DeepONet [Chen & Chen, 1995] guarantees that the branch network can approximate any continuous coefficient map $u \mapsto c_j$ from L^2 to $\mathbb{R}$, but for our specific problem the map is linear: the Fourier coefficient $c_j = \langle g, \mathcal{T}_j \rangle_{L^2}$ is a linear functional of the input u, and the branch network implements a linear functional via a single linear layer (the last hidden layer of the branch net). In the infinite-width limit, this linear approximation is exact, so the DeepONet effectively performs optimal Fourier coefficient extraction followed by Fourier synthesis. The worst-case bound in Theorem does not exploit this linearity, giving a pessimistic bias

term $p^{-2\beta/d}$ that holds for arbitrary nonlinear operators. For the linear operators in our experiments, the actual approximation error is dominated by the coefficient estimation error from finite data, not the structural bias, which explains why the empirical exponent exceeds the theoretical one.

2. **Regularity mismatch.** Our analytical test functions are infinitely differentiable (they are finite Fourier series with finitely many nonzero harmonics), placing them in $C^\infty(\Omega)$ and therefore in every Sobolev space $H^r(\Omega)$ for any $r \geq 0$. Assumption only requires $\mathcal{G}^\dagger(u) \in H^{s+\beta}(\Omega)$, a significantly weaker condition. The Sobolev embedding theorem states that $H^{s+\beta}(\Omega)$ embeds continuously into $C^{0,\gamma}(\Omega)$ for $\gamma < \beta - d/2$ when $\beta > d/2$, so the infinitely differentiable solutions in our experiments satisfy the embedding with room to spare. For smooth periodic functions, the Fourier coefficients decay faster than any polynomial rate: if $g \in C^\infty([0,1]^d)$, then $|c_k| = O(|k|^{-M})$ for every $M > 0$. This super-polynomial decay means the trunk tail $\sum_{j=p+1}^{\infty} c_j^2$ decays faster than any power of p, so the bias term $p^{-2\beta/d}$ is a severe overestimate. In the finite-p regime of our experiments, the actual bias is essentially negligible compared to the estimation error, and the MSE is dominated by the variance term, which decays as N^{-1} in the N-dependent complexity terms. This pushes the effective exponent toward 1.00 or higher, consistent with the observed superconvergence.

3. **Implicit regularization.** The Adam optimizer with fixed learning rate $\eta = 10^{-3}$ induces an implicit bias toward flat minima in the loss landscape, a phenomenon documented in the literature on deep learning [**?**]. For operator learning problems with Fourier-structured data, this bias manifests as a preference for learning low-frequency components before high-frequency ones. To see why, consider that the loss landscape of a DeepONet is highly non-convex, but the basins corresponding to low-frequency approximations are typically wider and have smaller loss barriers, making them more accessible to Adam's stochastic gradient steps. The spectral bias of gradient descent — which favors low frequencies in the early training phases — has been rigorously analyzed for deep ReLU networks in Yarotsky [2017], Petersen & Voigtlaender [2018] and observed empirically in vision models. For our PDEs, the dominant error modes are precisely the low-frequency Fourier components of the solution; if Adam learns these first, the error decays faster than the worst-case rate, explaining the superconvergence. The connection between Adam's implicit regularization and spectral prioritization is an active research area, and our experiments provide concrete evidence of this phenomenon in the operator learning setting.

A rigorous analysis of superconvergence is beyond the scope of this paper; we report it as an empirical observation. The honest claim is that the theoretical risk exponent ($N^{-1.00}$, i.e. $2\alpha_{\text{theor}} = 1.00$) and the observed empirical MSE exponents ($\hat{\gamma} = 0.93$ for Heat 1D, $\hat{\gamma} = 1.61$ for Poisson 2D) are all power laws with empirically verified monotonic decay.

5 Conclusion

We established a minimax upper bound for DeepONet (Theorem) learning PDE solution operators in Sobolev spaces. The theory predicts a risk exponent $2\alpha_{\text{theor}} = 2\beta/(d+\beta) = 1.00$ for both Heat 1D and Poisson 2D ($\alpha_{\text{theor}} = 0.50$). Empirical validation with single-run power-law fits shows clear convergence (MSE decreasing 7–26$\times$ across $N \in \{50, 400\}$),

with fitted MSE exponents $\hat{\gamma} = 0.93$ (Heat 1D, close to the theoretical 1.00) and $\hat{\gamma} = 1.61$ (Poisson 2D, substantially exceeding 1.00).

The paper makes *focused, auditable claims*: Theorem is an upper bound (not optimality), the lower bound is explicitly labeled a sketch, and the empirical rate is reported with the caveat that it is a single-run measurement.

The superconvergence finding ($\hat{\gamma} > 2\alpha_{\text{theor}}$) has significant implications for neural operator learning theory. It suggests that the current minimax upper bounds are pessimistic for structured PDEs with smooth solutions, and that future theoretical work should exploit additional structure — such as the linearity of the solution operator with respect to the input, the sinusoidal structure of the Fourier basis, or the implicit regularization of adaptive optimizers — to tighten the rates. The gap between theory ($\alpha = 0.50$) and practice ($\hat{\gamma} = 1.61$ for Poisson 2D) indicates that there is substantial room for refined approximation-theoretic analyses that go beyond the worst-case Sobolev regularity assumption. In particular, characterizing the function spaces for which DeepONet achieves rates faster than $N^{-1.00}$ would be a major advance.

For practitioners, the practical implication is that DeepONet is a reliable choice for learning smooth PDE solution operators, and that modest training set sizes ($N \sim 100$–400) are sufficient to achieve meaningful error reductions in many cases. The theory predicts that the convergence rate degrades as the PDE dimension d increases and improves as the PDE regularity gain β increases, which is consistent with practical experience: high-dimensional problems (e.g., Kolmogorov equations in finance, high-dimensional stochastic PDEs) will require larger training sets and possibly architecture modifications (such as adapted trunk bases or dimension-reduction techniques) to achieve comparable accuracy. The superconvergence observed for Poisson 2D suggests that for elliptic problems in moderate dimensions, DeepONet may be more efficient than the worst-case theory indicates, making it an attractive surrogate model choice for uncertainty quantification workflows where many forward evaluations are needed.

Not covered in this paper (open problems):

- **Full minimax lower bound with rigorous proof.** The sketch in Section 3.4 establishes thecorrect exponent $\beta/(d+\beta)$ but omits technical details. A complete proof requires constructing an L^2-packing of $\mathcal{G}(s,\beta)$ with the $e^{c \cdot p \log(d/p)}$ scaling and applying either Le Cam's method or Fano's method with controlled metric entropy [Yu et al., 1997, Wainwright, 2019]; see Bouchard et al. [2023] for a similar construction. This is a non-trivial extension that must carefully handle the interplay between the trunk basis truncation and the covering number of the resulting function space.

- **FNO spectral convergence theory and experiments.** Fourier Neural Operators use learned spectral filters rather than fixed basis functions, which complicates the bias analysis because there is no predetermined truncation error. The spectral convergence results in Kovachki et al. [2021], Lanthaler et al. [2024] apply under holomorphic extension assumptions that are strong but not always satisfied in practice. A minimax framework for FNO analogous to Theorem would require bounding the effective covering number of the FNO hypothesis space in a frequency-dependent norm, which is technically challenging. Empirically validating FNO convergence rates against analytical PDE solutions would complement the DeepONet experiments in this paper.

- **PAC-Bayes non-vacuous bounds with physics-informed priors.** The Massart-based complexity bound in Theorem gives a valid but potentially loose bound on the expected risk. PAC-Bayes theory [Kovačević et al., 2023] offers a principled way to incorporate prior knowledge about the PDE structure (e.g., the spectral decay of Fourier coefficients) into the bound, potentially yielding tighter non-vacuous guarantees. A physics-informed prior that assigns higher likelihood to low-frequency Fourier modes could substantially improve the bound's sharpness for the PDEs studied here.

- **Extension to discontinuous or non-smooth PDE solutions.** Many practically relevant PDEs — traffic flow, shock formation, interface problems — produce solutions with discontinuities or low regularity ($\beta < d/2$). The current theory relies on Assumption , which fails when the target function is not in $H^{s+\beta}$ for $\beta > d/2$. Extending the convergence theory to these settings requires replacing the Fourier coefficient decay assumption with a different approximation framework (e.g., wavelets, adaptive mesh refinement, or shock-capturing architectures) and would connect to the literature on neural network approximation of piecewise-smooth functions [Petersen & Voigtlaender, 2018, Kutyniok et al., 2022].

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